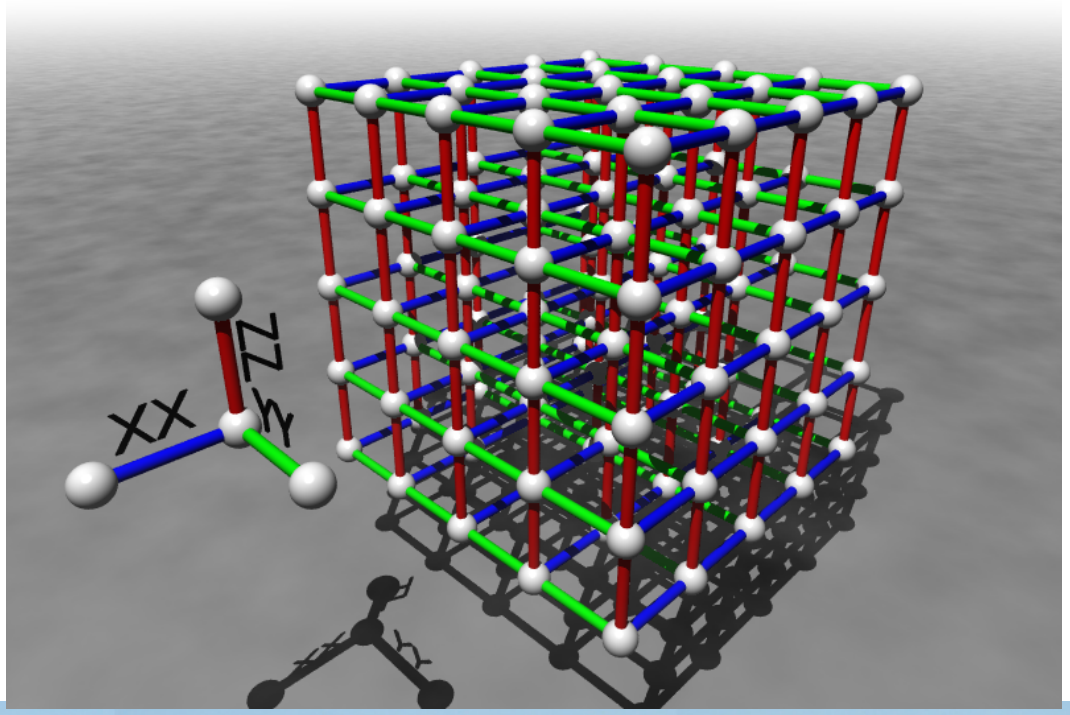
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Self-Correcting Quantum Computers

Dave Bacon



funded by

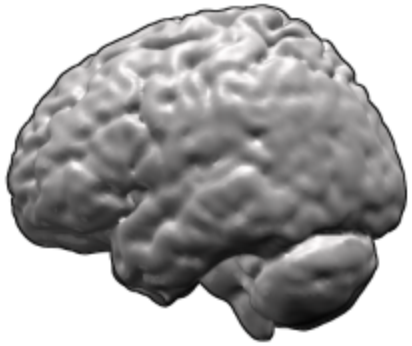


Department of Computer Science & Engineering
University of Washington

My Mortal Enemy



Success Stories



Life



Computers

How is it that nearly error free behavior can arise from systems whose components are unreliable?

$$|\psi\rangle \longrightarrow$$

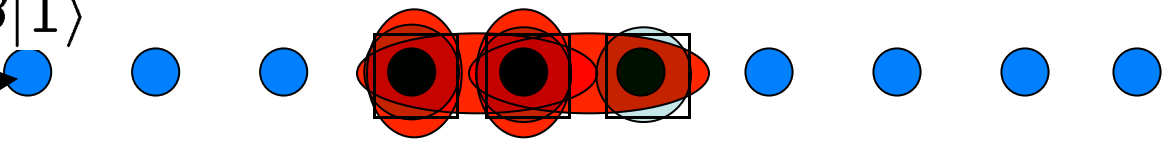


Robust Quantum Computers?

Dreamboat Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement

0

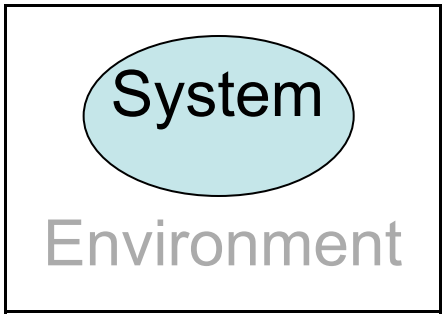
1

1

Obstructions



DECOHERENCE



Total Hamiltonian

$$H = H_S \otimes I + I \otimes H_E + H_I$$

system environment interaction

open quantum systems can become
entangled with their environment

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

coherent superposition incoherent sum

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

~~FAILS~~

More Obstructions



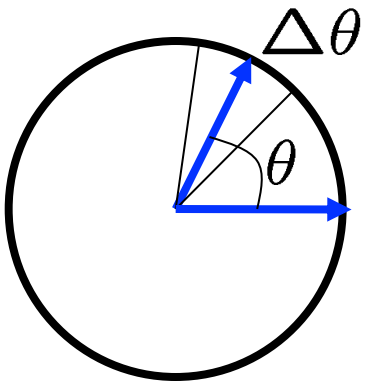
FAULTY CONTROL

System | Controller

Imprecise or faulty unitary gates

FAILS

H



Imperfect state preparation

FAILS

$|0\rangle$

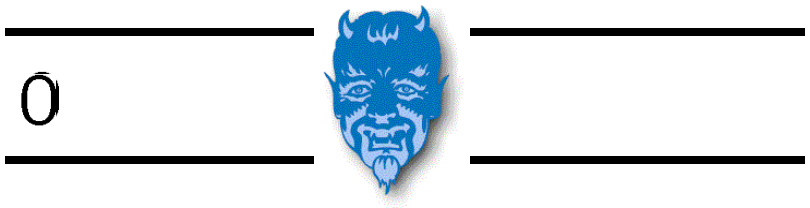
$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements

FAILS

$$\alpha|0\rangle + \beta|1\rangle \rightarrow \begin{cases} Pr(0) = |\alpha|^2 - \epsilon \\ Pr(1) = |\beta|^2 + \epsilon \end{cases}$$

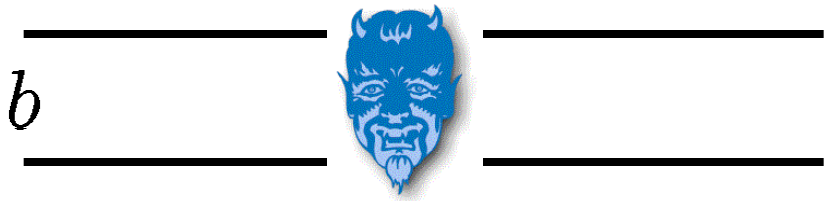
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

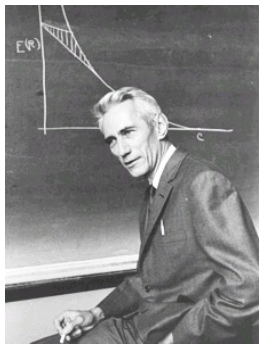
$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix} 000 \\ 001 \\ 010 \\ 100 \end{matrix}\right\} \rightarrow 0$$

$$\left.\begin{matrix} 111 \\ 110 \\ 101 \\ 011 \end{matrix}\right\} \rightarrow 1$$

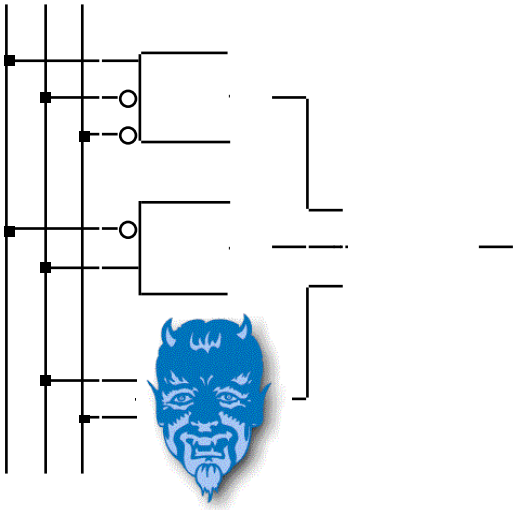
Probability of failure =*3(1-p)p2+p3*
=*3p2-2p3*
<*p* if *p*<*1/2*



Use redundancy to protect information against noise

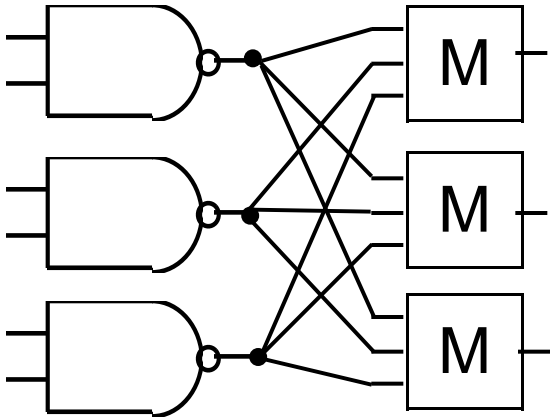
Claude Shannon 1948
“A mathematical theory of communication”

More Classical Solutions



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction

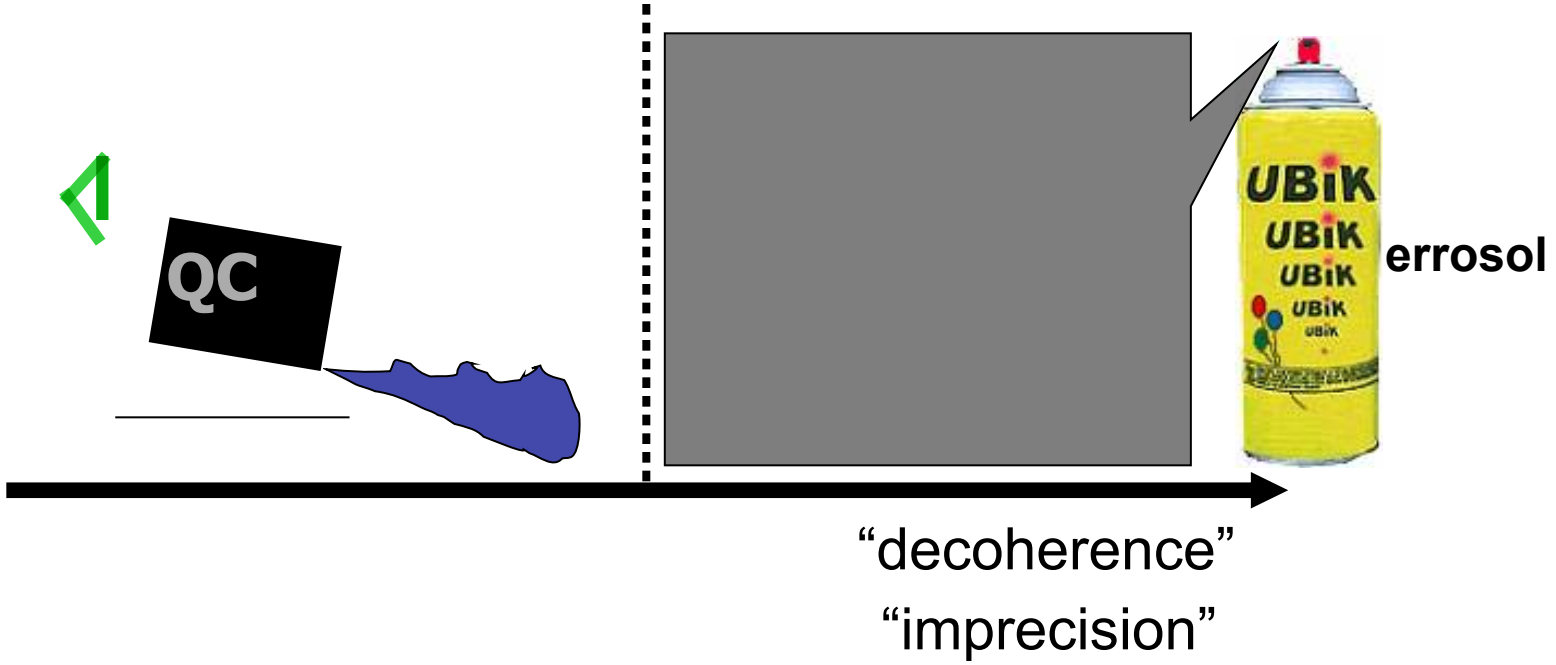




John von Neumann, 1956

“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

Wherefore Are Thou Quantum Computation?

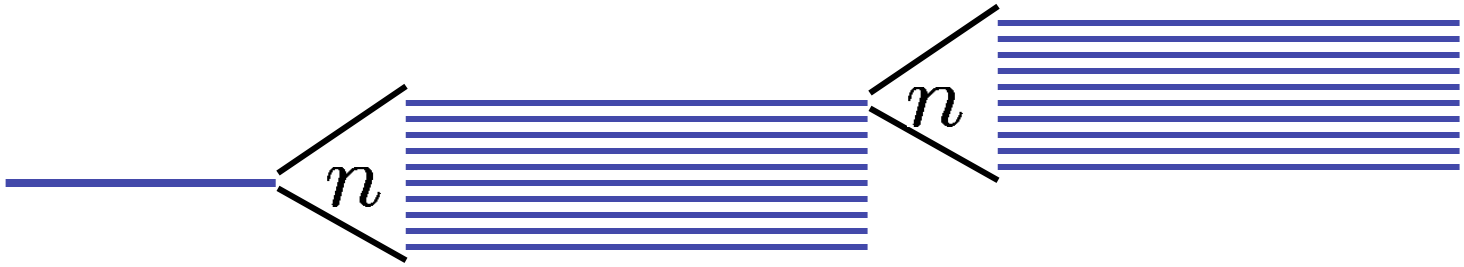


Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the one over the error

Perfection Through Concatenation



failure probability

p *cp2* *c(cp2)2*

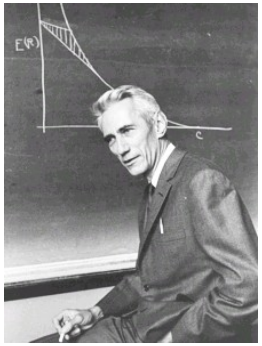
At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}}\left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log\frac{1}{p_{fail}}\right)$$

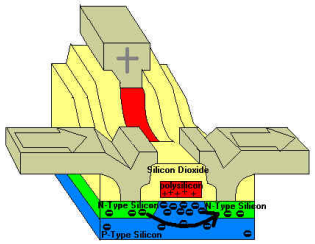
Back To Reality





theory



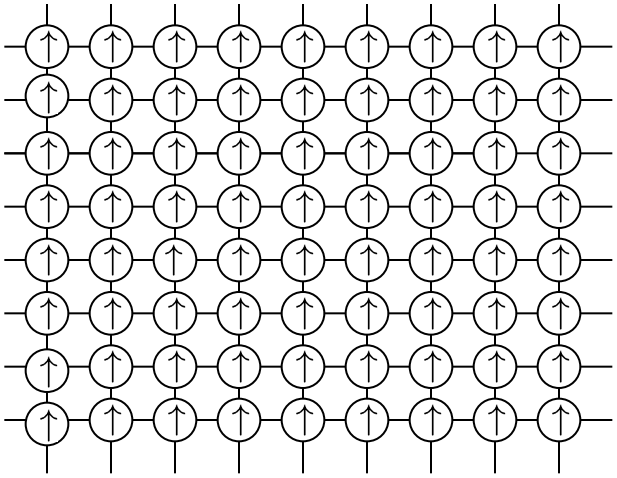


practice

Practical devices of information processing enact the ideas of fault-tolerant classical computation in the *physics* of the devices.

Toy Model



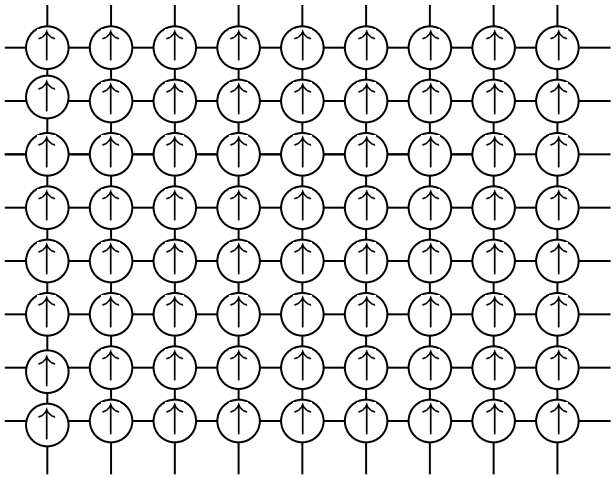


Ising model in two+ dimensio…

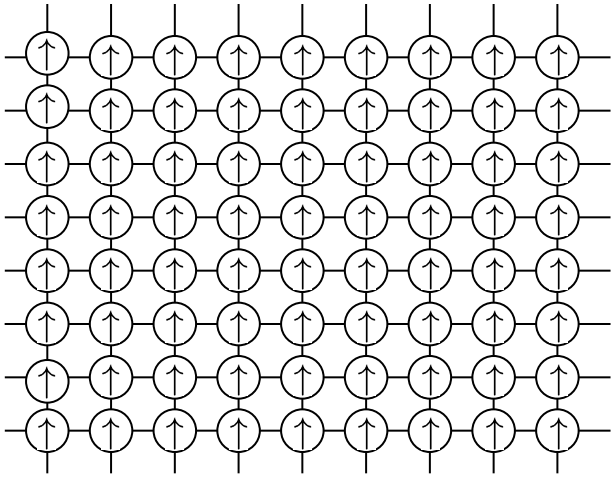
$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

Redundancy code:

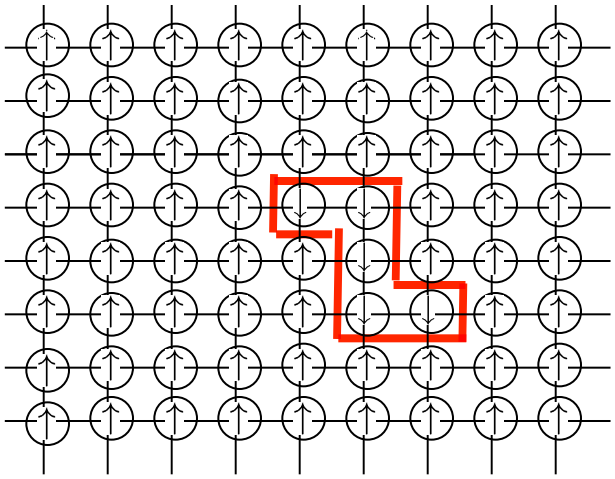
$0 \rightarrow$



$1 \rightarrow$

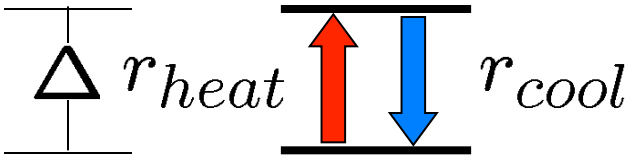


Dynamical Ising Model



Energy of excitation=
2J (Sum of perimeters of flipped domains)

Dynamics obeying principle of detailed balance:



$$\frac{r_{heat}}{r_{cool}} = e^{-\frac{\Delta}{k_B T}}$$

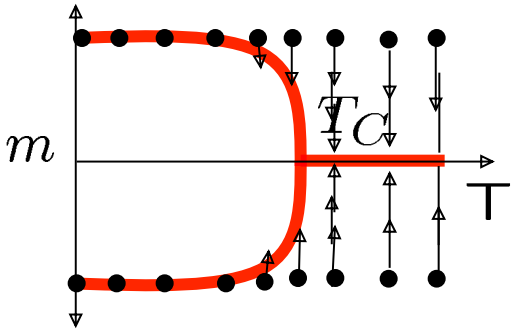
Competition between growth and decimation of flipped domains



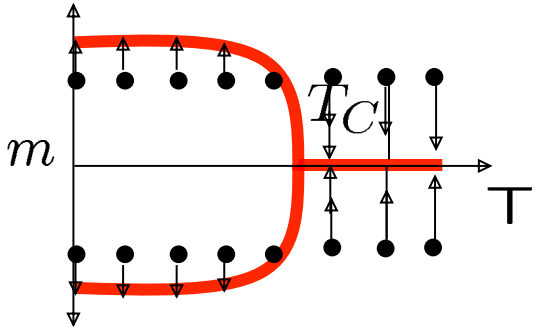
Robust



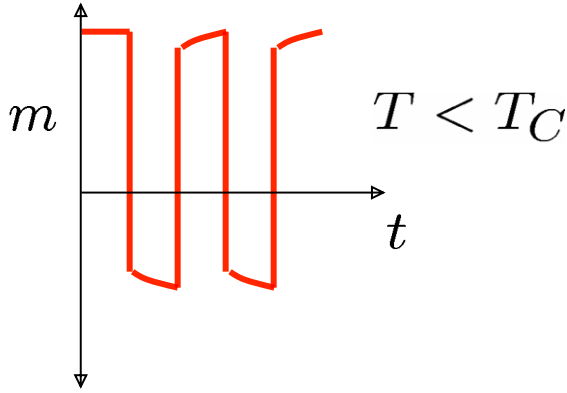
Relaxation to thermal



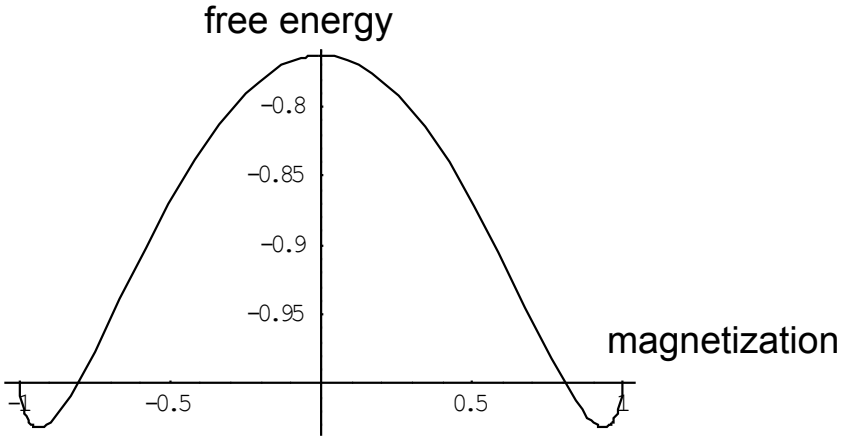
Imperfect preparation



Manipulation

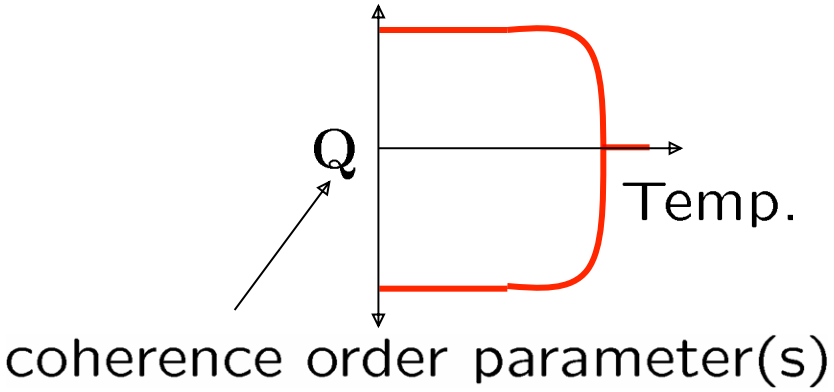


The two encoded states have lifetimes which grow with the size of the system:



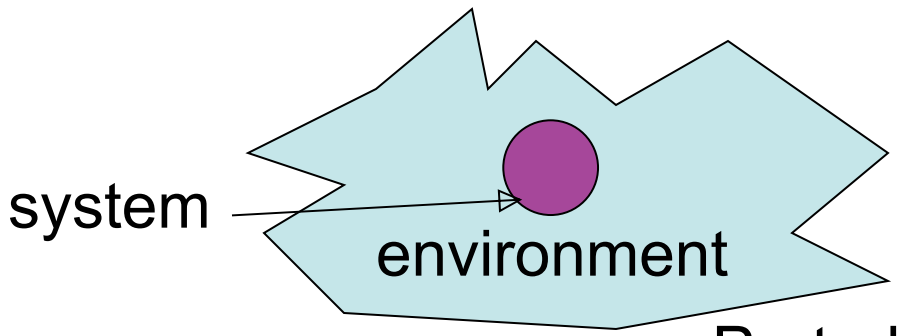
Quantum Hard Drive?

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?

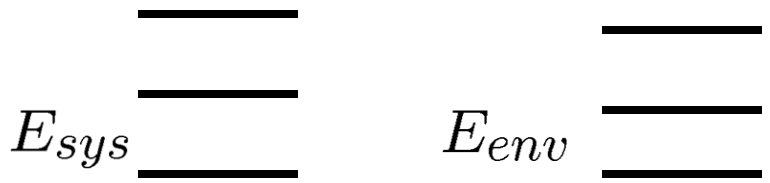


Today in QC: we have theory (like von Neumann, Shannon) but not devices which implement these idea (without severe device complexity overhead.)

Energetics of Decoherence

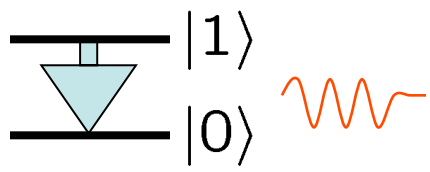


$$H = H_{sys} + H_{int} + H_{env}$$



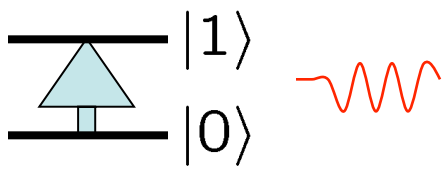
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



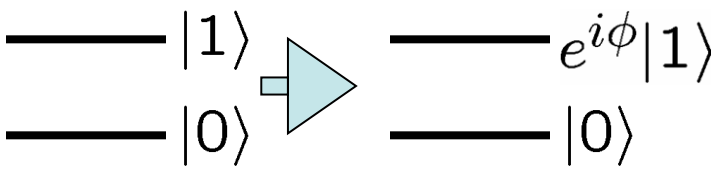
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$

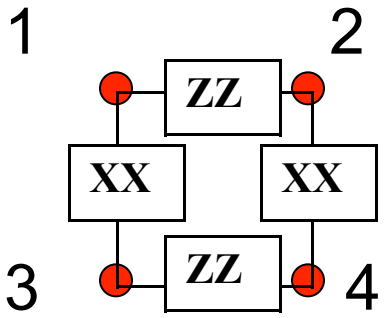


dephasing

$\Delta E_{sys} = 0$

by **cooling environment** we can eliminate heating

1st Attempt



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

Quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$
$$S_2 = X_1X_2X_3X_4$$
$$S_i|\psi_C\rangle = |\psi_C\rangle$$
$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$
$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

Code is a symmetry of the Hamiltonian:

$$[H, S_1] = [H, S_2] = 0$$

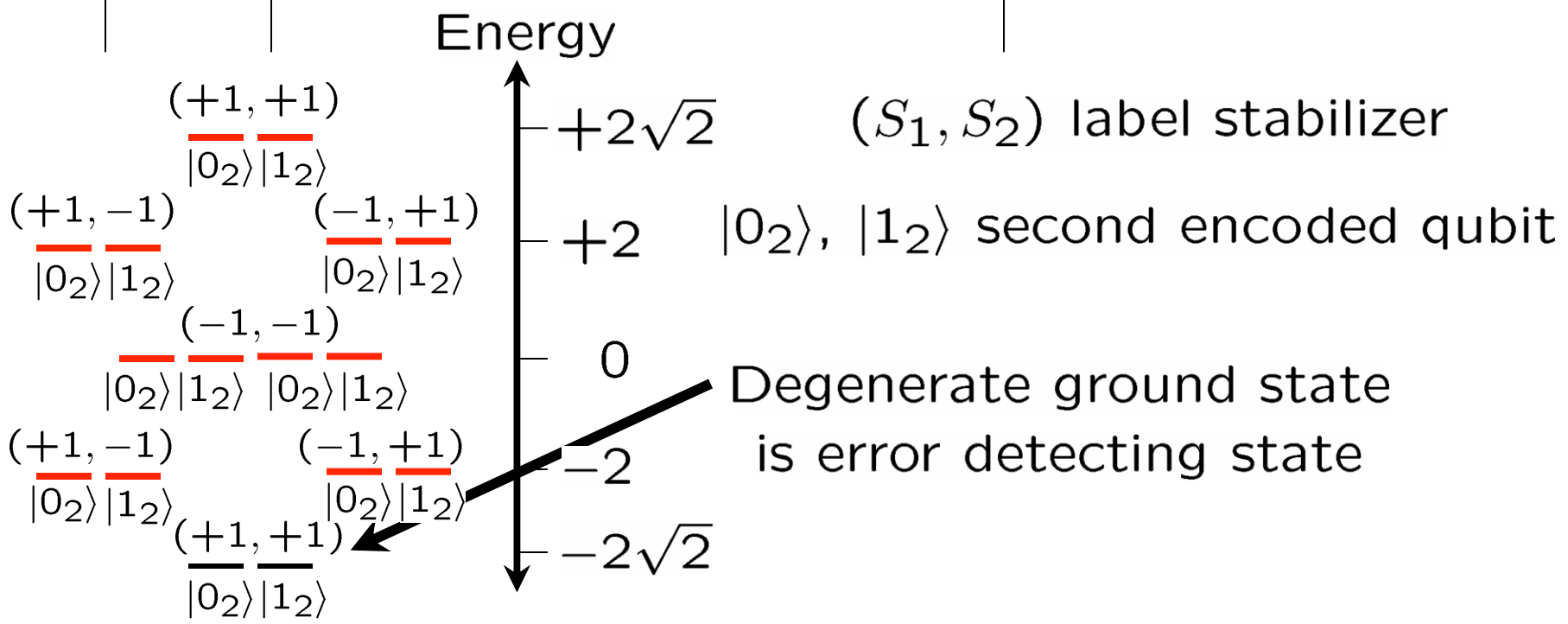
Code detects any single qubit error:

$$\{P, S_i\} = 0 \qquad P \text{ e.g. } Q \otimes I \otimes I \otimes I$$

H does not depend on second qubit (use to store encoded qubit)

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0

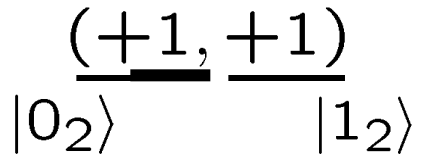


ground state is +1 eigenvalue of four qubit interactions

Supercoherence

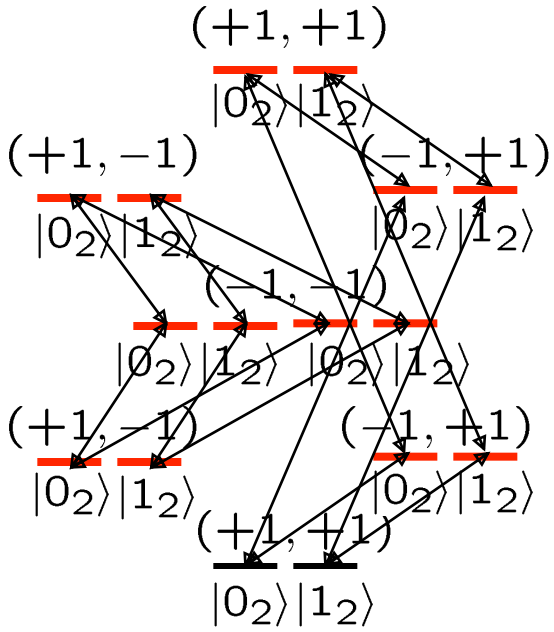
ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.



✗ **FORBIDDEN**

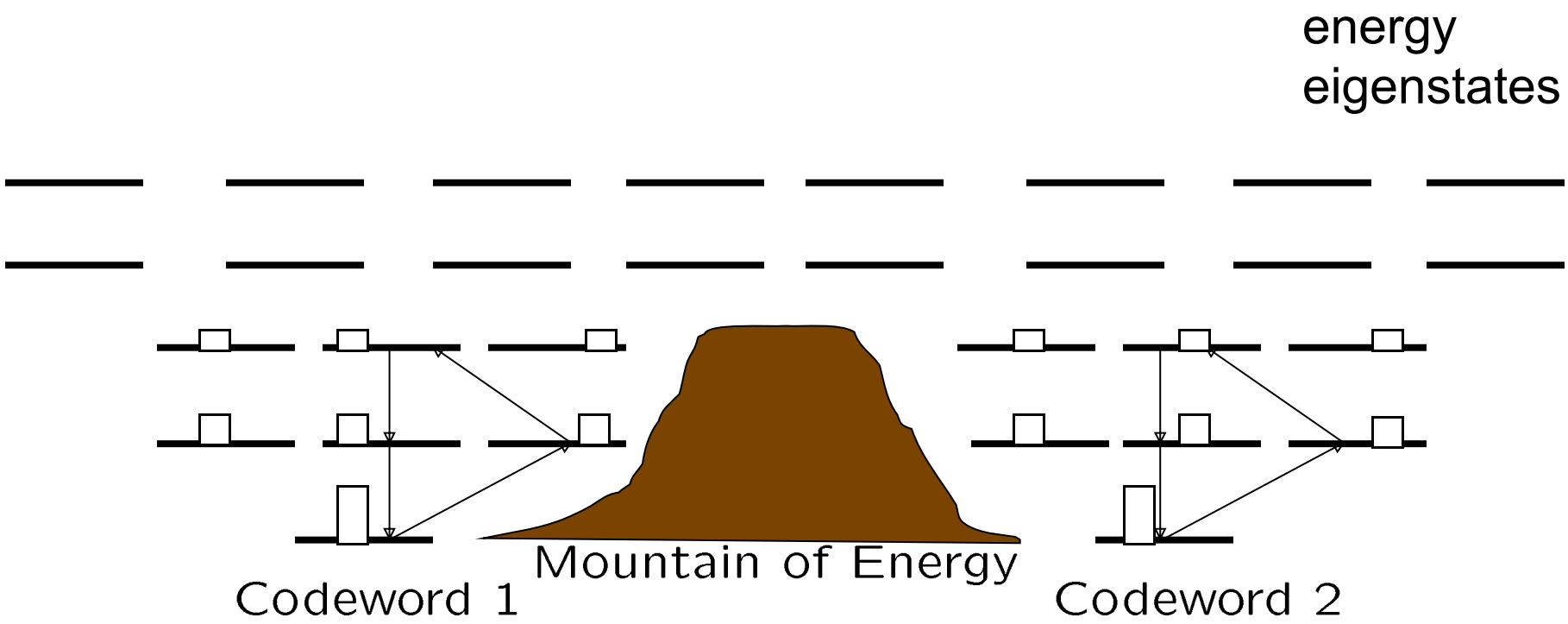
Effect of **ZIII**



To first order all single qubit decoherence becomes heating

Energy gap and low temperature suppress 1st order decoherence

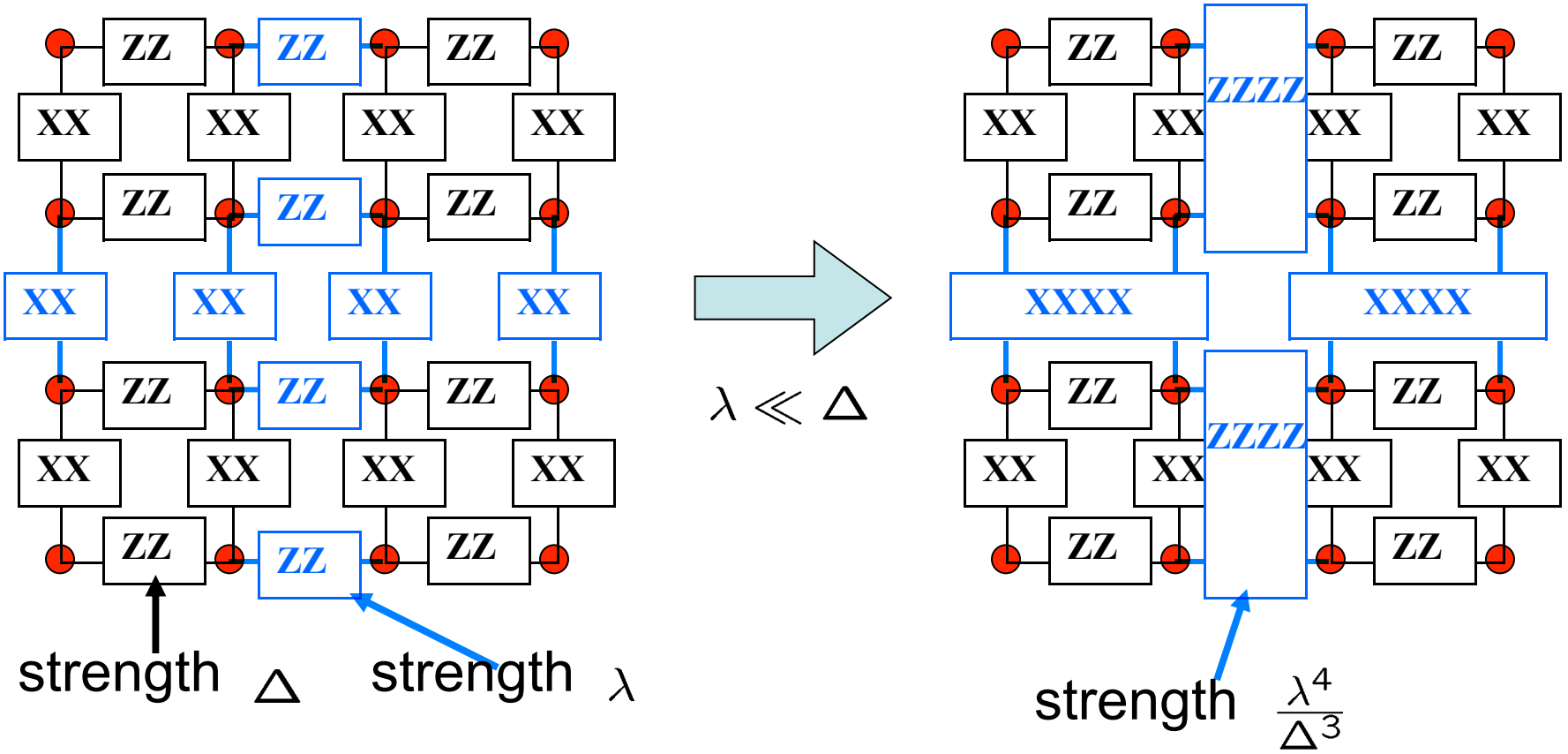
Self-Correction

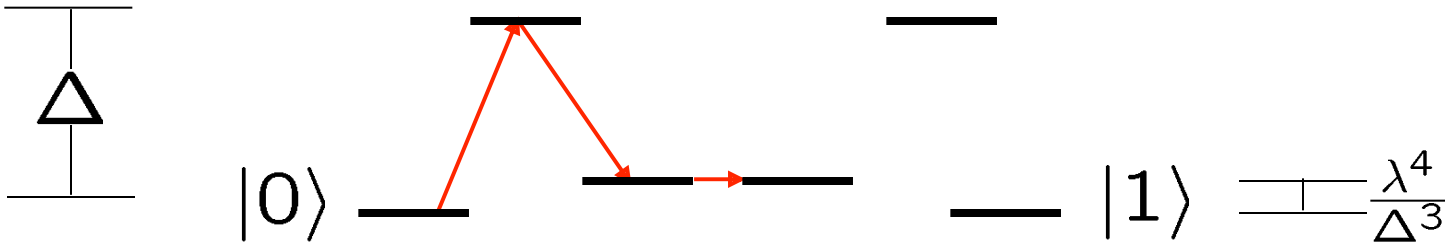


Important for high T operation as well as for fault-tolerance

Energy gap is not enough to guarantee fault-tolerance
(though it is sufficient for memory)

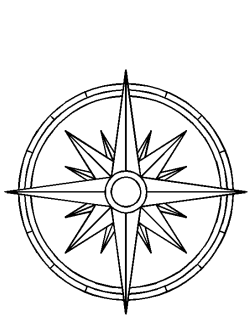
Concatenate?

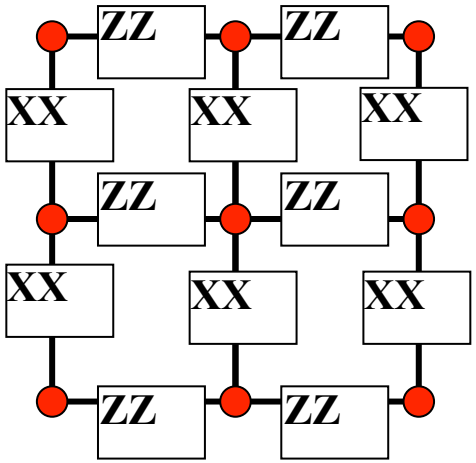


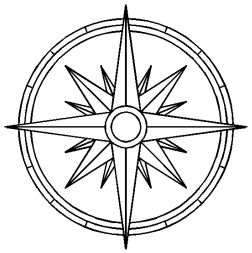


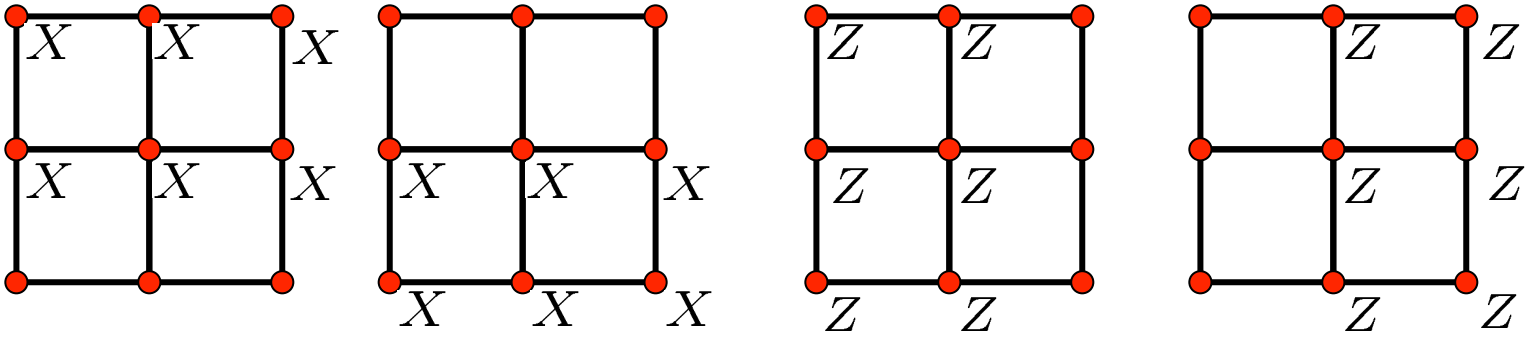
Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$









Abelian symmetries

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two qubits

$$\mathcal{H}_2 \otimes \mathcal{H}_2$$

<u>encode a qubit in a subspace</u>

$$|0_L\rangle = |1\rangle \otimes |1\rangle$$

$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$\alpha|0_L\rangle + \beta|1_L\rangle$$

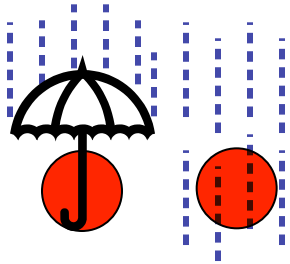
<u>encode a qubit in a subsystem</u>

$$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$$

generic subsystem:

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

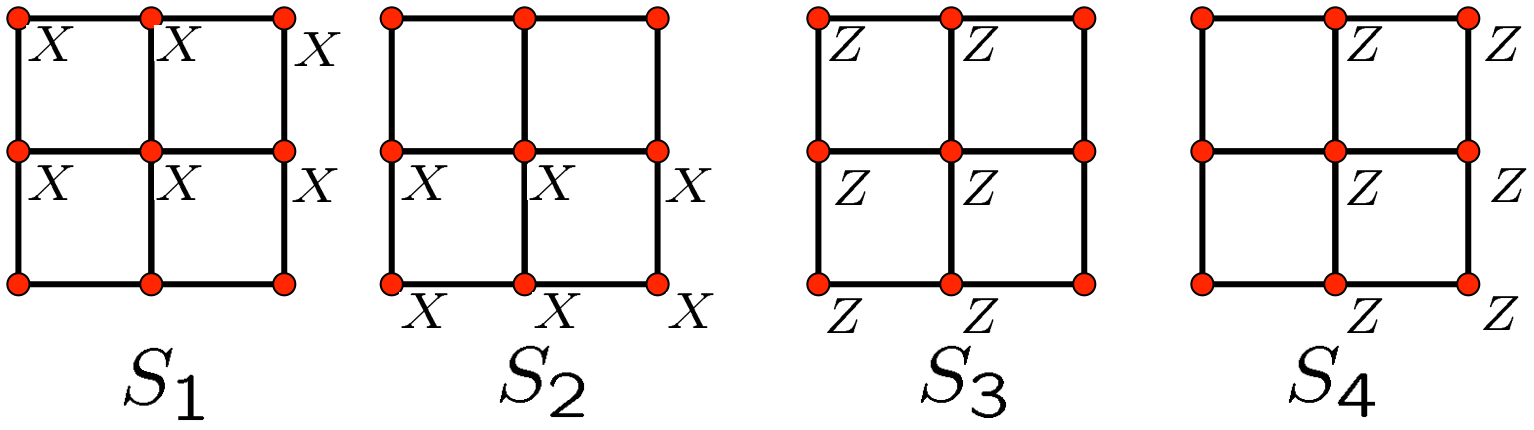
$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0$ — $\mathcal{E}$ — $\mathcal{R}$ — $(|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$

error recovery

encoded quantum information

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Quantum Subsystem Code



. Measure these operators

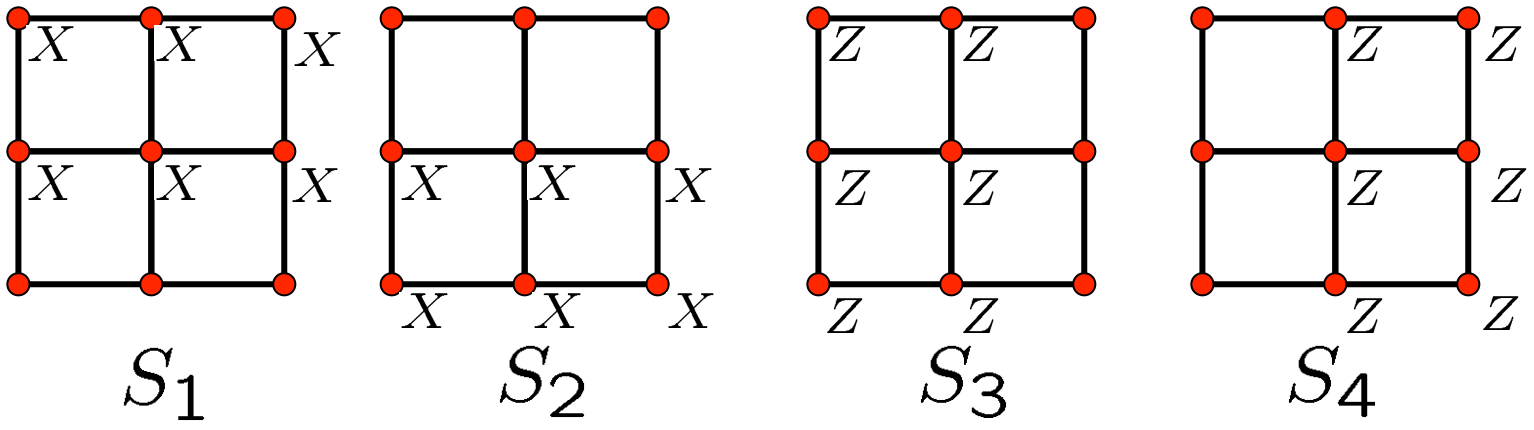
. Diagnose parity of bit and phase flips in column and rows

3. Apply recovery routine for parity of errors

Routine fixes error UPTO an operation generated by bonds of Hamiltonian

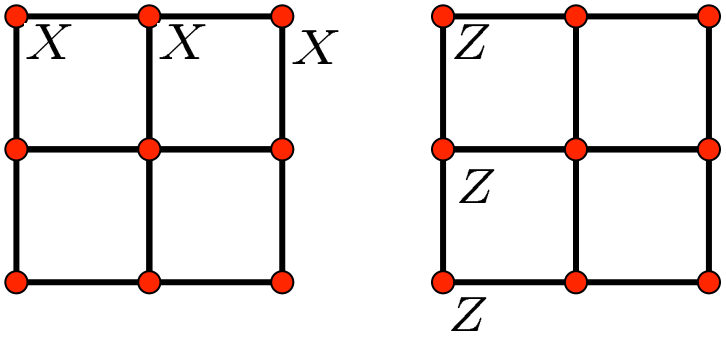
$$X_{i,j}X_{i+1,j}, Z_{i,j}Z_{i,j+1}$$

Quantum Subsystem Code

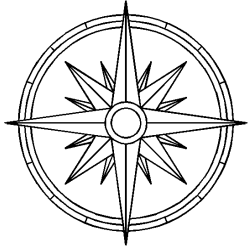


$$X_{i,j}X_{i+1,j},\, Z_{i,j}Z_{i,j+1}$$

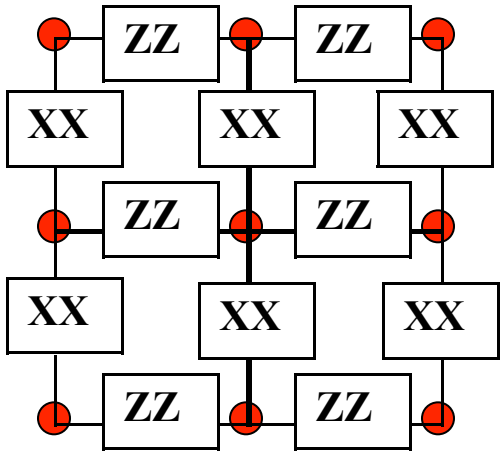
extra symmetry which implies states are doubly degenerate:



Quantum Compass Model



$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

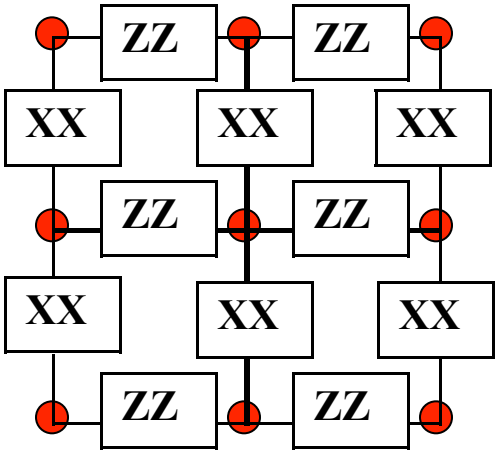


Ground state is a new kind of quantum error correcting code

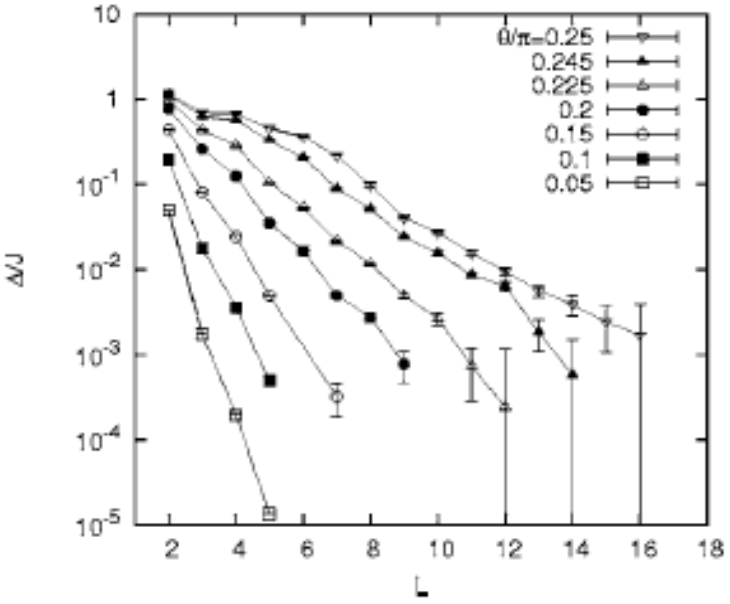
(This code is related to Shor's original code as well as Kitaev's toric code and has remarkable fault-tolerant properties [Aliferis unpublished])

quantum error correcting subsystem

Self-Correcting?



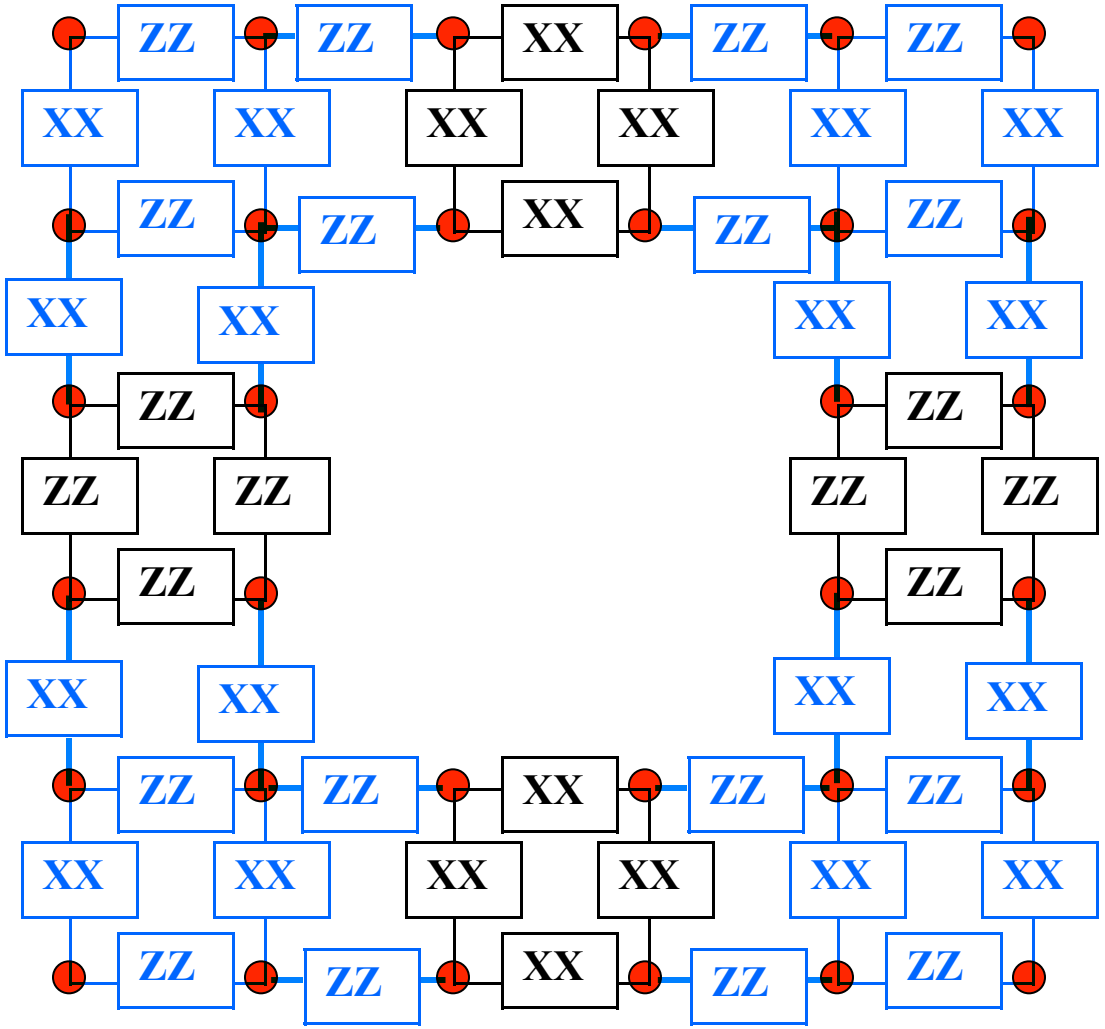
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Energy gap vanishes for large lattice size Λ

[Dourier, Becca, Mila 2005]

The Proper Concatenation



Self-correcting!

Towards Robust Qubits

Wire many faulty qubits together into a fault-tolerant scheme



Engineer physical systems whose physics enacts error correction

